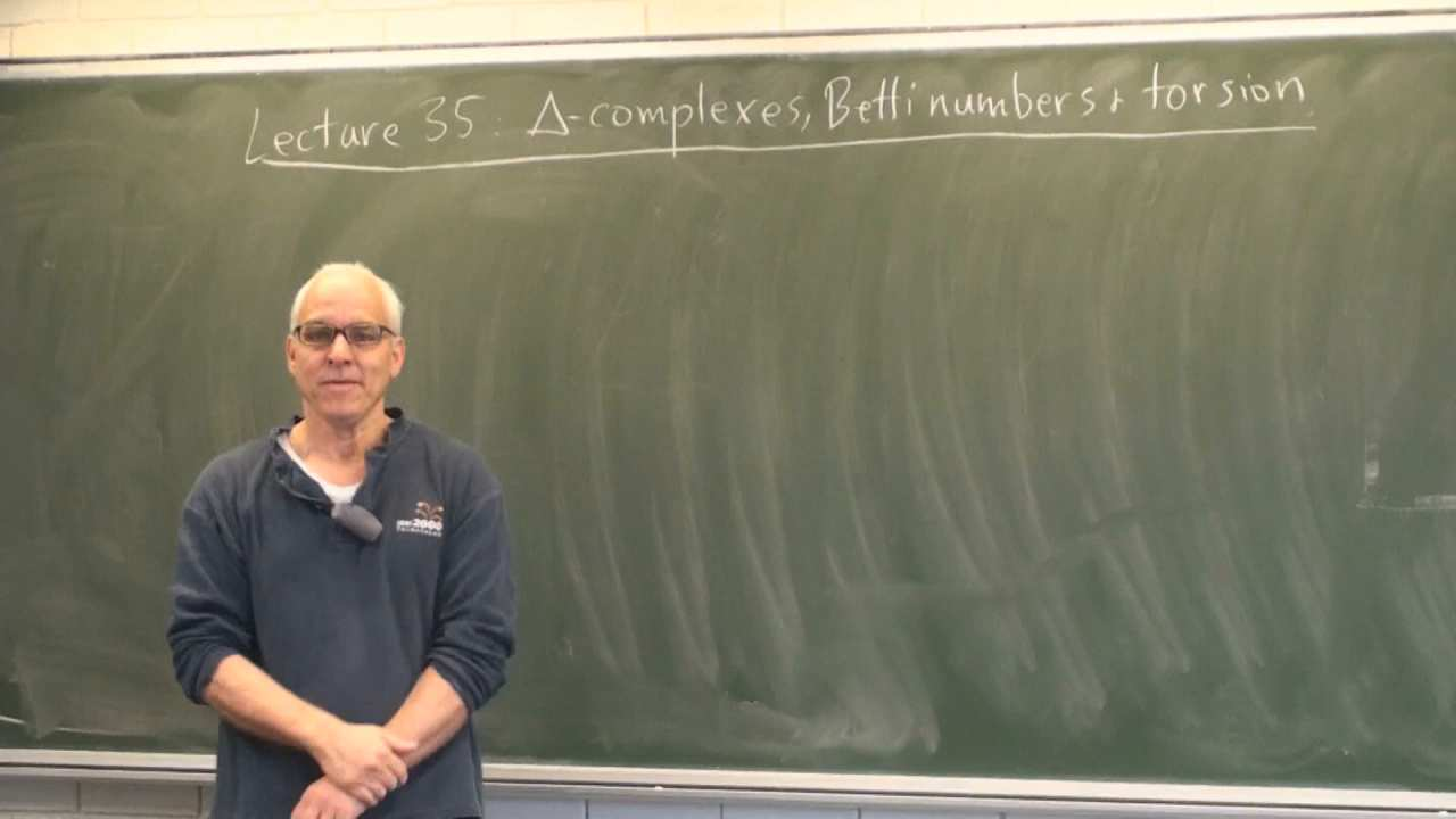
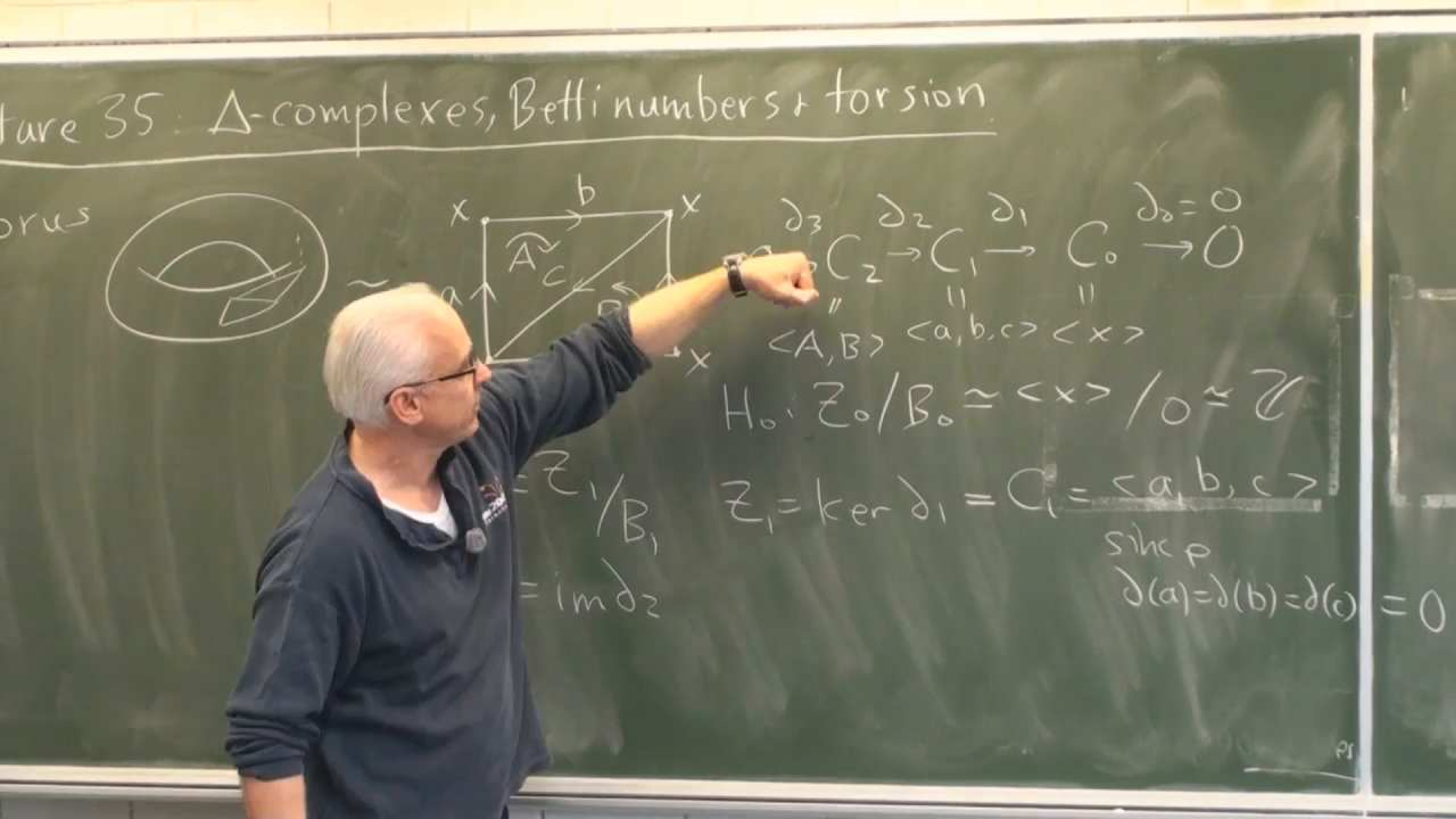
Algebraic Topology: a beginner's course

This lecture:

AlgTop35: Delta complexes, Betti numbers and torsion


Lecture 35: Δ-complexes, Betti numbers + torsion



ture 35: Δ-complexes, Betti numbers + torsion
orus
x
b
x
A
C
B
a
x
$\partial_3 \quad \partial_2 \quad \partial_1 \quad \partial_0 = 0$
$\to C_2 \to C_1 \to C_0 \to 0$
$\langle A, B \rangle \quad \langle a, b, c \rangle \quad \langle x \rangle$
$H_0 : Z_0 / B_0 \simeq \langle x \rangle / 0 \simeq \mathbb{Z}$
$= Z_1 / B_1$
$= \operatorname{im} \partial_2$
$Z_1 = \ker \partial_1 = C_1 = \langle a, b, c \rangle$
since
$\partial(a) = \partial(b) = \partial(c) = 0$

subgps of $\mathbb{Z}$: $n\mathbb{Z}$, $n \in \mathbb{Z}$.

$(3\mathbb{Z}, 5\mathbb{Z}, \dots)$.

has 1-dim subgps: $\langle \vec{v} \rangle = n\vec{v}$, $n \in \mathbb{Z}$.

$\simeq \mathbb{Z}$

has 2-dim subgps: $\langle \vec{u}, \vec{v} \rangle$: $n\vec{u} + m\vec{v}$, n, m

H

fundamental domain for H. H

cosets of H in $G = \mathbb{Z} \oplus \mathbb{Z} \longleftrightarrow$ pts inside

i.e. $G/H \simeq$ pts in

index H in G is 2.

subgps of $\mathbb{Z}$: $n\mathbb{Z}$, $n \in \mathbb{Z}$

($3\mathbb{Z}$, $5\mathbb{Z}$, ...)

has 1-dim subgps: $\langle \vec{v} \rangle = n\vec{v}$, $n \in \mathbb{Z}$

$\simeq \mathbb{Z}$

has 2-dim subgps: $\langle \vec{u}, \vec{v} \rangle$: $n\vec{u} + m\vec{v}$, $n, m \in$

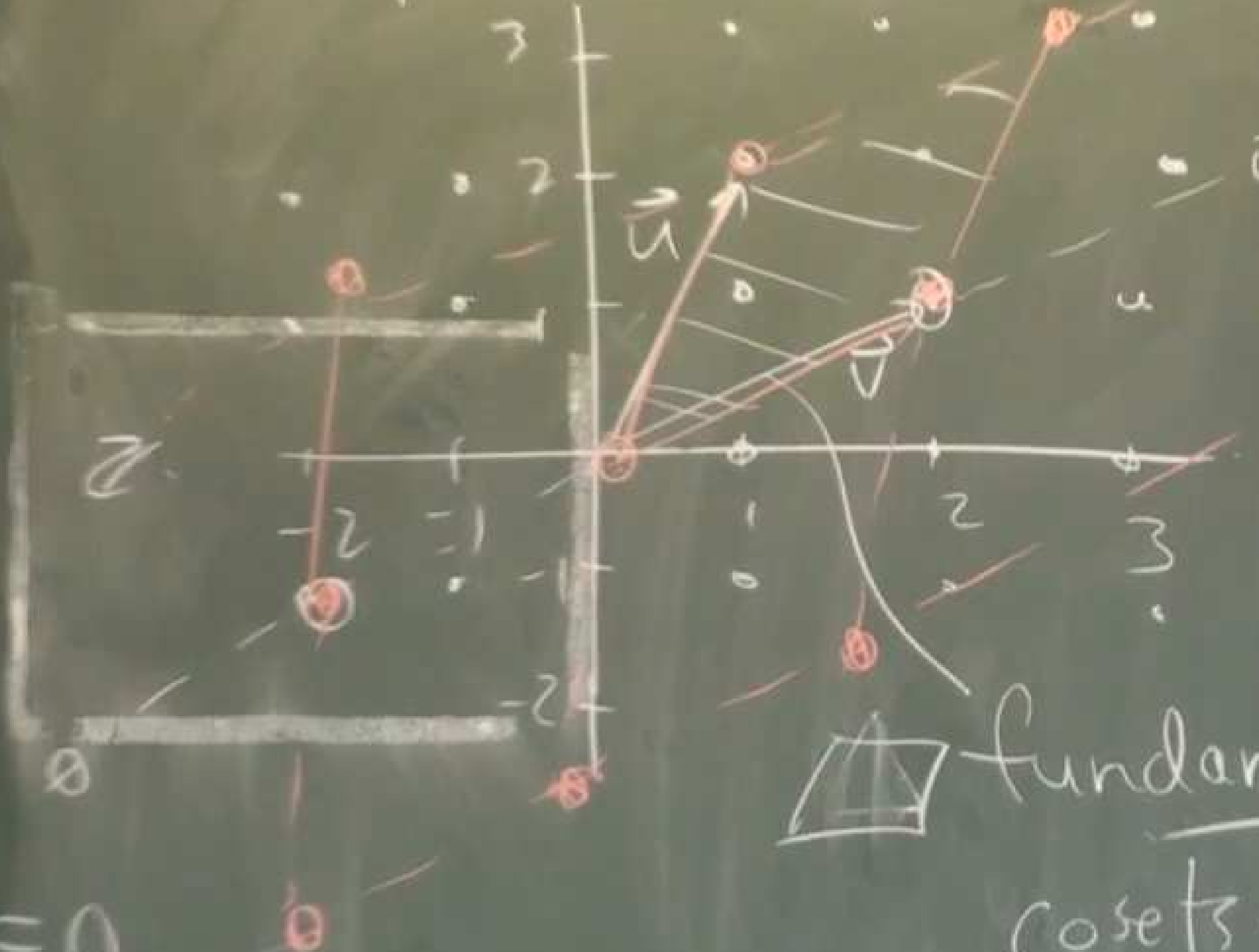
<u>fundamental domain</u> for H. H

cosets of H in $G = \mathbb{Z} \oplus \mathbb{Z}$ $\longleftrightarrow$ pts inside

ie G/H ... pts in

index H in G is 2.

Subgps of $\mathbb{Z}\oplus\mathbb{Z}$:



subgps of $\mathbb{Z}$: $n\mathbb{Z}$, $n\in\mathbb{Z}$

$(3\mathbb{Z}, 5\mathbb{Z}, \ldots$

has 1-dim subgps: $<\vec{v}$

$\simeq \mathbb{Z}$

has 2-dim subgps: $<\vec{u}$

H

fundamental domain for H.

cosets of H in $G=\mathbb{Z}\oplus\mathbb{Z} \leftrightarrow$ pts ins

ie $G/H \simeq$ pts in $\square$

(c) $=0$.

$H_2 = Z_2 / B_2$ $\quad Z_2$: $\ker \partial_2$: $\alpha A + \beta B$

$$\boxed{\begin{aligned} H_0(T) &= \mathbb{Z} \\ H_1(T) &= \mathbb{Z} \oplus \mathbb{Z} \\ H_2(T) &= \mathbb{Z} \\ H_n(T) &= 0 \quad n \geq 3 \end{aligned}}$$

$$\alpha A + \beta B \xrightarrow{\partial} \alpha(a+b+c) + \beta(a+b+c)$$

$$= (\alpha+\beta)(a+b+c) = 0$$

$$\iff \alpha = -\beta$$

$$\therefore \ker \partial_2 = \langle A - B \rangle.$$

$B_2 = \operatorname{im} \partial_3 = 0$ so $H_2 = \langle A - B \rangle / 0 \cong \mathbb{Z}$

$H_2 = Z_2 / B_2$ $\quad Z_2 = \ker \partial_2 \ni \alpha A + \beta B$

$H_2 = Z_2/B_2$

$\downarrow S$

$\boxed{\begin{array}{l} H_0(T) = \mathbb{Z} \\ \cancel{H_1(T) = \mathbb{Z}}\oplus\mathbb{Z} \\ H_2(T) = \mathbb{Z} \\ H_n(T) = 0 \quad n \geq 3 \end{array}}$

$\alpha(a+b+c) + \beta(a+b+c)$

$= (\alpha+\beta)(a+b+c) = 0$

$\iff \alpha = -\beta$

$\therefore \ker \partial_2 = \langle A - B \rangle$

$B_2 = \operatorname{im} \partial_3 = 0$ so $H_2 = \langle A-B \rangle / 0 \cong \mathbb{Z}$

$= \langle a, b, c, \langle\rangle\rangle / \langle a+b+c, 2c \rangle \cong \dfrac{\mathbb{Z}\oplus\mathbb{Z}}{2\mathbb{Z}\oplus\mathbb{Z}} \cong \mathbb{Z}/2\mathbb{Z} = \mathbb{Z}_2 = \{0, 1\}$